the transition toward portable, hand-launched, miniaturized drone systems. Designed primarily for tactical reconnaissance, surveillance, and target acquisition, the Raven became one of the most widely used small unmanned aerial vehicles in the world, revolutionizing battlefield intelligence and situational awareness at the unit level. Unlike its predecessors, which required complex launch systems or runways, the RQ-11 could be deployed by a single operator in the field, making it an essential tool for ground troops, Special Forces, and border patrol units.

The RQ-11 Raven features a lightweight airframe constructed from composite materials and a wingspan of just 1.4 meters, allowing it to be carried and launched by hand. It is powered by a quiet electric motor, providing an operational endurance of approximately 60–90 minutes and a range of up to 10 kilometers. The drone is equipped with modular payloads, including electro-optical and infrared (EO/IR) cameras, which transmit live video and telemetry data to a portable ground control station (GCS) in real time. Its GPS-based autopilot system allows for fully autonomous flight, including pre-programed routes, waypoint navigation, and automatic return-to-base functions (Van Bourgondien, 2012).

Operationally, the RQ-11 Raven became a critical asset during the Iraq and Afghanistan conflicts, where it provided real-time reconnaissance and surveillance for platoon-level operations. It was used for monitoring enemy movement, identifying ambush sites, and assessing terrain before troop advancement. The drone's low acoustic signature and lightweight design made it ideal for stealthy reconnaissance missions in complex terrains such as urban environments and mountainous regions.

The Raven's user-friendly interface and modular system design allowed rapid deployment and easy maintenance, contributing to its widespread adoption by over 30 military forces worldwide, as well as by civil defense,